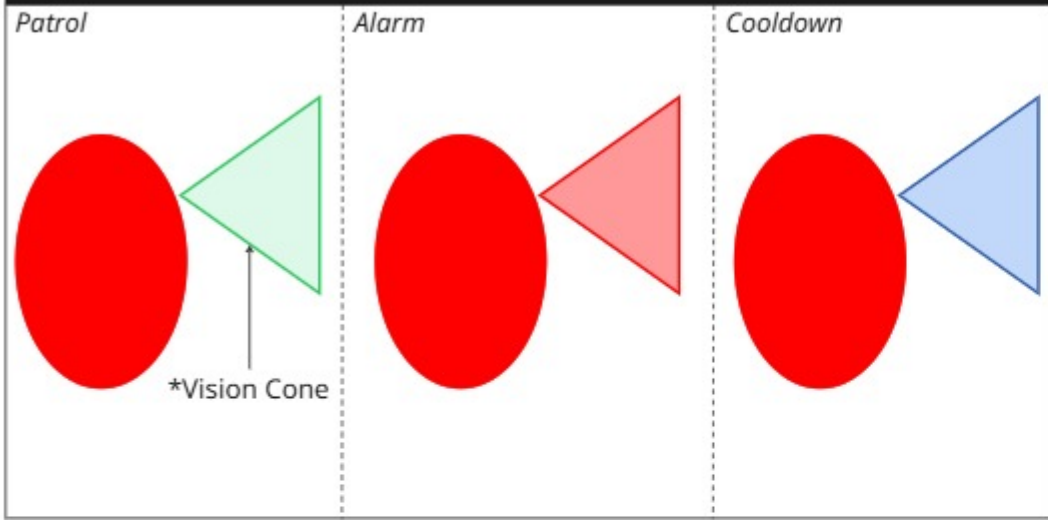
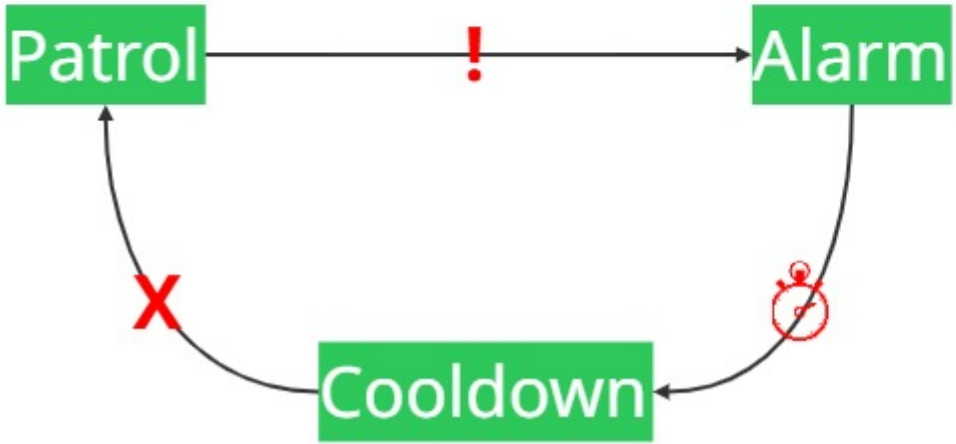
During the development on this project, I didn't face many challenges. The only notable one was with my Alarm AI. It was programmed to follow a route around the level until it sees the Player. Once seen, it goes into the "Alarm" state and signals the other AI. After 10 seconds, the Alarm AI will switch to a "Cooldown" state, where it can not activate an alarm until it reaches one of the many patrol points in the level. This "Cooldown" state should switch once it reaches a patrol point. But there was a bug that when the AI reaches a patrol point, it will stay in the "Cooldown" state and go onto the next patrol point; It would go on continuously like that. So, I fixed the code so that once the Alarm AI reaches any point, it leaves the "Cooldown" state and successfully goes back to patrolling. Now after it begins patrolling again, it can set off the alarm if it sees the Player, instead of being stuck in cooldown and not being able to activate its alarm anymore.

YouTube Video Link: https://youtu.be/DNepMi09lCY

Alarm AI

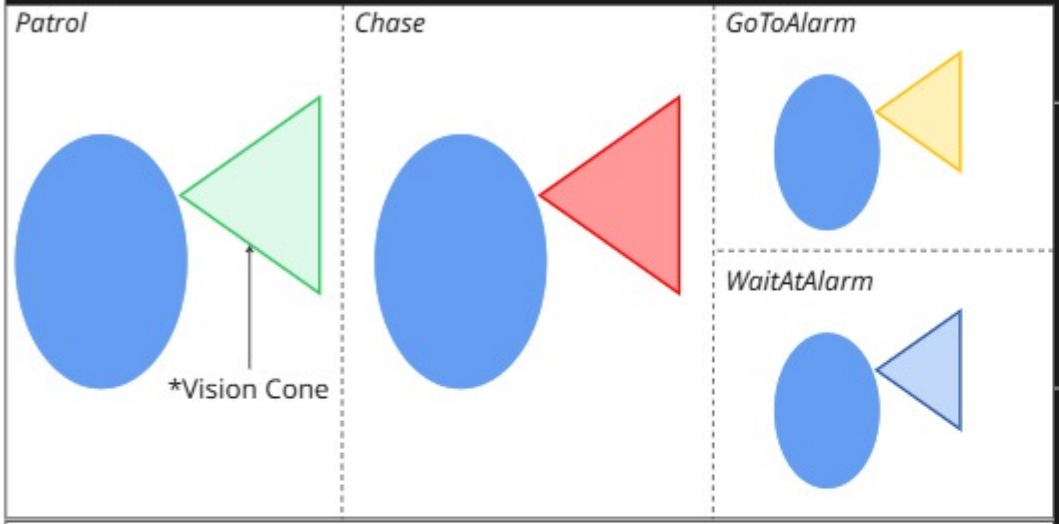


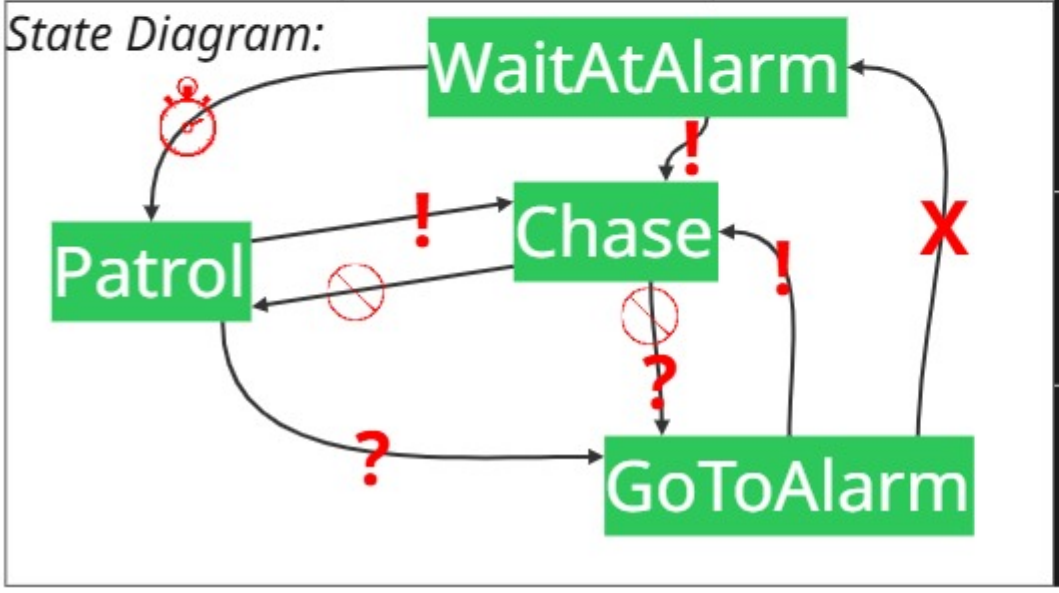
State Diagram:



Properties	• It's purpose is only to alert other enemies of the player location • It roams all over the level	• Stays in place when alarm goes off • It uses a vision cone to see objects and the player
State Behaviors	• Patrol: This behavior has the AI roaming a set of points until triggered to another state • Alarm: This has the AI blaring an alarm in place, signaling nearby enemies	• Cooldown: This is a state where the AI makes its way back to the patrol route, and can't set off the alarm while in this state
State Triggers	• ! This trigger is when the AI sees the player • X This trigger is when the AI reaches its next patrol point	• This trigger signifies a time limit before switching states
AI - Player Interactions	• Upon seeing the player, using it's vision cone, they will sound their alarm • They will stay in the spot they were last when they saw the player	• This AI enemy will patrol across the entire level until they find the player
AI - AI Interactions	• When this enemy's alarm goes off, it signals all the other "Normal" enemies to it's location	

Patrol AI





Properties	• It's purpose is to guard the exit door • It roams between two patrol points	• Is hostile towards the Player • It uses a vision cone to see
State Behaviors	• Patrol: This behavior has the AI roaming a set of points until triggered to another state • Chase: Follows the Player while they are in sight	• GoToAlarm: When the alarm goes off, this AI will make its way to the location of said alarm • WaitAtAlarm: Once arrived at the alarm location, wait there for a few seconds
State Triggers	• ! This trigger is when the AI sees the player • X This trigger is when the AI reaches the alarm location • ? This signifies when this AI is alerted by an alarm	• This trigger signifies a time limit before switching states • This signifies when the AI can't see the Player anymore
AI - Player Interactions	• Upon seeing the Player, using it's vision cone, they will move towards them	• If the AI gets close enough to the Player, it will kill them
AI - AI Interactions	• When the Alarm AI signals an alarm, the Patrol AI will respond to it by moving to its location	